Kenneth Breugelmans

Data Scientist · MSC PSA European Terminal · Antwerp, Belgium

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PROFILE

I've spent years on the floor of one of Europe's busiest container terminals — operating quay cranes, driving straddle carriers, managing live cargo flows. I became a data scientist because I saw what better information could do for operations that still run largely on instinct and experience.

Today I work at the intersection of heavy industry and modern data engineering: building integration layers for legacy terminal operating systems, designing digital twins that make real-time operations legible, and implementing reinforcement learning models for container optimisation. I don't do mystery — you'll know what I'm building, why, and where it stands.

EXPERIENCE

Data Scientist — Jul 2024 – Present

MSC PSA European Terminal (MPET) · Antwerp

Building ML pipelines and RL models for container optimisation. Designed and implemented a production-grade integration layer connecting the terminal's legacy TOS to ML-driven decision pipelines — enabling automated operational actions without modifying the core system. Building a real-time digital twin for operational decision-making and monitoring.

Ship-to-Shore Crane Operator — Aug 2019 – Jul 2024

MSC PSA European Terminal (MPET) · Antwerp

Operated single and dual hoist cranes on some of the world's largest container vessels. Collaborated with technical teams on complex software and hardware troubleshooting. Completed a full-time Data Science programme (Turing College) in parallel with this role.

Straddle Carrier Operator — Aug 2017 – Aug 2019

MSC PSA European Terminal (MPET) · Antwerp

Container transport and yard logistics across complex terminal layouts. Responsible for equipment reliability checks and coordination with maintenance teams.

Dock Worker — Sep 2015 – Jul 2024

CEPA · Antwerp

Container preparation, discharge coordination, and crane guidance under limited visibility conditions. Specialised in twistlock troubleshooting and live operational problem-solving.

Production Operator

Mar 2013 – Sep 2015

Xtraflex · Lier

Custom Teflon hose fabrication to tight tolerances. Managed the Trax racing store — building bespoke brake lines for classic cars, rally cars, and racing motorcycles.

Operator (seasonal)

Summers 2010 & 2011

Atlas Copco Airpower · Antwerp

Installation of custom electrical modules into industrial compressors according to client specifications.

EDUCATION

Data Science — Professional Programme

May 2023 – Sep 2024

Turing College · Online

Intensive programme covering machine learning, data engineering, deep learning, and applied statistics. Completed while working full-time as a crane operator on rotating shift patterns.

TECHNICAL SKILLS

Languages	Python · SQL · JavaScript · C# · Java · Kotlin · Mojo · GCODE
Data & Pipelines	Pandas · Polars · DuckDB · Dask · NumPy · Scikit-Learn · OpenCV
ML & Modelling	XGBoost · LightGBM · CatBoost · FastAI · PyTorch · Lightning AI · Optuna · SHAP
RL & Simulation	Unity3D · ML-Agents · PPO · SAC
Visualisation	Matplotlib · Plotly · Seaborn · Folium · Tableau · Power BI · Deck.gl
Web & Publishing	Flask · Streamlit · Vue · HTML/Bootstrap · Quarto
Cloud	Microsoft Azure · Google Cloud · Amazon AWS
CAD / Engineering	Autodesk Fusion 360 · Autodesk Inventor · Onshape · SketchUp

SELECTED PROJECTS

Point-in-Time Data Reconstruction

2024 – Present

MSC PSA European Terminal (MPET) — Production

CDC-based pipeline capturing every live mutation to critical TOS tables and streaming them to the cloud via Qlik — enabling full point-in-time reconstruction of any table's state, down to the second. The TOS's only native option was periodic snapshots, leaving large gaps whenever retroactive analysis or incident investigation required knowing exactly what the data looked like during a specific operation. The pipeline replaced that blind spot with a complete, queryable change history that persists indefinitely.

Predicted Discharge Time

2024 – Present

MSC PSA European Terminal (MPET) — Production

Predictive model estimating container discharge time during active vessel operations. Truckers historically waited until a ship had fully departed before planning pickup — with no reliable signal on when individual containers would be accessible. The model gives a discharge window hours in advance, letting hauliers plan their arrival before the quay clears. Achieved 84% of predictions within one hour of actual discharge time.

Digital Twin - Proof of Concept 2024 – Present

MSC PSA European Terminal (MPET) — Production

Web-based real-time digital twin of the terminal's live operational state — ingesting data from the TOS and equipment systems to render yard occupancy, vessel positions, crane assignments, and equipment movements as they happen. Giving planners and shift supervisors a single coherent view of what is actually happening on the terminal, replacing the patchwork of screens and radio calls that operational awareness previously depended on.

Operational Integration Layer 2024 – Present

MSC PSA European Terminal (MPET) — Production

Production-grade integration layer bridging a legacy TOS with ML model pipelines. Enables automated operational decisions at scale with full roll-back capability. Built with Python, C#, REST APIs, and event-driven architecture.

RL Quay Crane 2021

Personal R&D — Proof of Concept

Custom Unity3D simulator and RL training environment to explore whether a reinforcement learning agent could replicate quay crane operator behaviour. Trained agents using PPO and SAC via ML-Agents. Concluded that RL can handle ~95% of crane tasks given sufficient compute and training time.

DOMAIN KNOWLEDGE

Over a decade of hands-on experience across the full operational hierarchy of a large container terminal — from dockworker to straddle carrier driver to quay crane operator. Deep understanding of the workflows, physical constraints, and failure modes that do not appear in any manual: berth planning, yard density management, equipment choreography under vessel deadlines, and the human factors that determine whether a shift runs smoothly or doesn't.

LANGUAGES

Dutch	Native
English	Professional proficiency (duolingo certified 130/160)
French	Basic